

Use Case Document

ReqAgent — AI Requirement Intake & Tracker

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1. Actors

Actor	Description
Stakeholder	Business user submitting a requirement (any department)
Business Analyst	Reviews, manages, and approves requirements
AI Processor	Groq LLM that generates user stories and classifications
System	The ReqAgent application

2. Use Cases

UC-01: Submit Business Requirement

Field	Details
Actor	Stakeholder
Trigger	Stakeholder has a business need to document
Precondition	Stakeholder has access to the web application
Postcondition	Requirement saved and AI analysis displayed

Main Flow:

1. Stakeholder navigates to the Submit Requirement tab
2. Stakeholder enters title, department, role, name, and business objective
3. Stakeholder enters detailed requirement description and optional notes
4. Stakeholder clicks Submit & Generate AI Analysis

5. System validates all required fields
6. System saves raw intake to MongoDB
7. System sends data to AI Processor
8. System displays user stories, MoSCoW, priority, and conflict analysis

Alternative Flow: At step 5, if required fields are missing, system displays error and halts submission

UC-02: Save Raw Intake to MongoDB

Field	Details
Actor	System
Trigger	Stakeholder submits requirement form
Precondition	Form validation passes
Postcondition	Document saved in MongoDB with unique ObjectId

Main Flow:

1. System collects all form fields into a document object
2. System appends submission timestamp and sets processed flag to False
3. System inserts document into MongoDB raw_intake collection
4. System returns MongoDB ObjectId for cross-referencing

UC-03: Generate User Stories

Field	Details
Actor	AI Processor
Trigger	Raw requirement saved to MongoDB
Precondition	Valid requirement text available
Postcondition	3 user stories with acceptance criteria returned

Main Flow:

1. System constructs prompt with requirement details and existing requirements list
2. System sends prompt to Groq API (llama-3.3-70b-versatile)
3. AI generates 3 user stories in standard format
4. AI generates 2-3 acceptance criteria per story in Given/When/Then format

5. System parses JSON response and returns structured story objects

Alternative Flow: System retries API call once on failure. If second attempt fails, system logs error.

UC-04: Classify MoSCoW

Field	Details
Actor	AI Processor
Trigger	Requirement text received for processing
Precondition	Requirement description is available
Postcondition	One MoSCoW category assigned

Main Flow:

1. AI reads requirement title, description, and business objective
2. AI evaluates business criticality and dependencies
3. AI assigns one of: Must Have / Should Have / Could Have / Won't Have
4. Classification returned as part of JSON response

UC-05: Assign Priority Score

Field	Details
Actor	System + AI Processor
Trigger	Requirement submitted with submitter role
Precondition	Submitter role captured in form
Postcondition	Priority (High/Medium/Low) assigned with justification

Main Flow:

1. System reads submitter role from form data
2. System applies role-weight mapping: C-Suite/VP → High, Manager/Lead → Medium, End User → Low
3. AI provides written justification for priority
4. Priority and justification stored in Postgres

UC-06: Detect Conflicts

Field	Details
Actor	AI Processor
Trigger	New requirement submitted
Precondition	At least one existing requirement in the system
Postcondition	Conflict report returned

Main Flow:

1. System retrieves all existing requirement titles and descriptions from MongoDB
2. System includes existing requirements in AI prompt
3. AI compares new requirement against all existing ones
4. AI identifies overlaps, contradictions, or scope conflicts
5. AI returns list of conflicts or 'No conflicts detected'
6. Conflicts stored in Postgres conflicts table

UC-07: Update Requirement Status

Field	Details
Actor	Business Analyst
Trigger	BA selects new status from dropdown
Precondition	Requirement exists in tracker
Postcondition	Status updated and audit log entry created

Main Flow:

1. BA selects new status from dropdown within requirement card
2. BA clicks Update button
3. System validates new status differs from current status
4. System updates status in requirements table
5. System inserts audit log entry with old status, new status, timestamp
6. System refreshes tracker view

Alternative Flow: If same status selected, system displays warning. No database changes made.

UC-08: Export Traceability CSV

Field	Details
Actor	Business Analyst
Trigger	BA clicks Download as CSV button
Precondition	Traceability matrix has data
Postcondition	CSV file downloaded to BA's device

Main Flow:

1. System converts current filtered matrix to CSV format
2. System encodes as UTF-8
3. Browser initiates file download
4. File saved as traceability_matrix.csv